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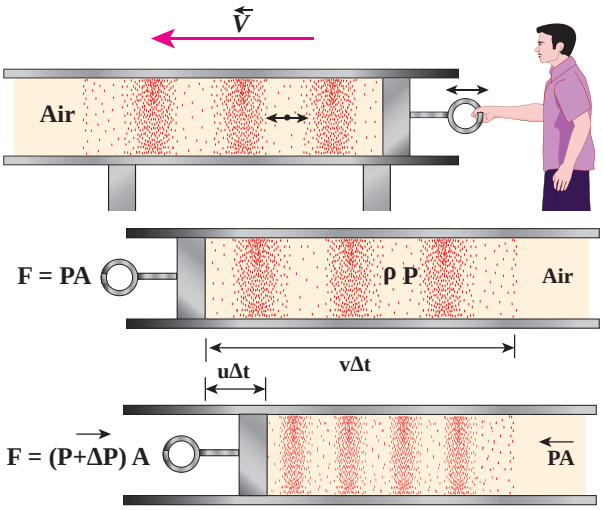


Figure 11.16 Longitudinal waves in the fluid by displacing the fluid using a piston

axis of the cylinder. Let ρ be the density of the fluid which is initially at rest. At $t = 0$, the piston at left end of the tube is set in motion toward the right with a speed u.

Let u be the velocity of the piston and v be the velocity of the elastic wave. In time interval Δt, the distance moved by the piston $\Delta d = u\ \Delta t$. Now, the distance moved by the elastic disturbance is $\Delta x = v\Delta t$. Let Δm be the mass of the air that has attained a velocity v in a time Δt. Therefore,

$$\Delta m = \rho\ A\ \Delta x = \rho\ A\ (v\ \Delta t)$$

Then, the momentum imparted due to motion of piston with velocity u is

$$\Delta p = [\rho\ A\ (v\ \Delta t)]u$$

But the change in momentum is impulse.

The net impulse is

$$I = (\Delta P\ A)\Delta t$$

Or $$(\Delta P\ A)\Delta t = [\rho\ A\ (v\ \Delta t)]u$$

$$\Delta P = \rho\ v\ u \qquad (11.14)$$

When the sound wave passes through air, the small volume element (ΔV) of the air undergoes regular compressions and rarefactions. So, the change in pressure can also be written as

$$\Delta P = B\frac{\Delta V}{V}$$

where, V is original volume and B is known as bulk modulus of the elastic medium.

But $V = A\ \Delta x = A\ v\ \Delta t$ and

$\Delta V = A\ \Delta d = A\ u\ \Delta t$

Therefore,

$$\Delta P = B\frac{A u\ \Delta t}{A v\ \Delta t} = B\frac{u}{v} \qquad (11.15)$$

Comparing equation (11.14) and equation (11.15), we get

$$\rho v u = B\frac{u}{v} \text{ or } v^2 = \frac{B}{\rho}$$

$$\Rightarrow v = \sqrt{\frac{B}{\rho}} \qquad (11.16)$$

In general, the velocity of a longitudinal wave in elastic medium is $v = \sqrt{\frac{E}{\rho}}$, where E is the modulus of elasticity of the medium.

Cases: For a solid :

(i) one dimension rod (1D)

$$v = \sqrt{\frac{Y}{\rho}} \qquad (11.17)$$

where Y is the Young's modulus of the material of the rod and ρ is the density of the rod. The 1D rod will have only Young's modulus.

(ii) Three dimension rod (3D) The speed of longitudinal wave in a solid is

$$v = \sqrt{\frac{K + \frac{4}{3}\eta}{\rho}} \qquad (11.18)$$

where η is the modulus of rigidity, K is the bulk modulus and ρ is the density of the rod.